Data-Driven Activity Recognition and Energy Consumption Forecasting for Heavy-Duty Vehicles

Thesis, Halmstad university

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February 9, 2025

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1 Introduction

Heavy-duty vehicles consume a substantial portion of their total energy through auxiliary systems, which significantly affects overall operational efficiency and vehicle range, particularly for battery electric vehicles (BEVs). Accurately modeling and forecasting auxiliary energy consumption is challenging due to its strong dependence on vehicle operational states, environmental conditions, and driver behavior. Existing approaches may provide limited predictive performance in complex real-world operations, as they often fail to capture the stochastic and highly variable nature of auxiliary energy usage.

Identifying vehicle operational activities—such as driving, idling, stopping, loading and unloading, and the usage of auxiliary systems like electric power take-off (ePTO)—is a critical component for effective fleet management, safety monitoring, and operational planning. While driving-related activities are directly linked to vehicle propulsion, ePTO usage represents a distinct class of operational behavior associated with auxiliary equipment operation rather than vehicle motion. Traditional activity recognition methods typically rely on GPS-based rules, handcrafted features, or time-based heuristics. These approaches are labor-intensive, difficult to generalize across vehicle types and operating environments, and often require large amounts of manually labeled data, limiting their scalability.

To tackle these challenges, this thesis develops an activity recognition framework based on multivariate sensor time-series data collected from the vehicle CAN bus. The model is trained offline using labeled data provided by domain experts, ensuring interpretability and accuracy of key activities. Once trained, the model performs real-time inference on streaming data, enabling the system to detect operational states such as driving, idling, or loading while maintaining robustness and adaptability. This approach reduces labeling effort while ensuring accurate and robust activity predictions.

Building on the recognized activities, this work further investigates auxiliary energy consumption forecasting using a learning paradigm that combines offline and online learning. Initially, offline models are trained on historical data to capture general consumption patterns. These models are subsequently adapted through online learning to account for evolving vehicle behavior, changing operating conditions, and seasonal variations, enabling continuous model improvement during deployment.

By integrating activity recognition with adaptive energy consumption forecasting, the aim of this study is to improve the operational efficiency of heavy-duty BEVs and support the development of intelligent, data-driven fleet management systems.

2 Problem Formulation

The problem addressed in this thesis is the development of a learning framework capable of recognizing vehicle operational activities and forecasting auxiliary energy consumption in heavy-duty battery electric vehicles under evolving operating conditions. Given multivariate time-series data collected from onboard CAN bus sensors, the system must identify distinct vehicle activities and estimate their associated auxiliary energy usage in a sequential data setting. New operational activities, such as those introduced by additional equipment or changing usage patterns, may appear over time. At the same

time, retraining models from scratch after each update is infeasible for large-scale fleet deployment. These constraints motivate a framework where activity recognition relies on offline self-supervised pretraining with real-time inference, while energy forecasting leverages a combination of offline training on historical data and online updates to adapt to evolving operational conditions.

To address these challenges, this research distinguishes between activity recognition and auxiliary energy forecasting. Activity recognition models are first trained offline in a self-supervised manner on multivariate time-series data to learn meaningful feature representations of vehicle behavior. A feature encoder f_θ maps each fixed-length segment $\mathbf{X}_i \in \mathbb{R}^{T \times S}$ to a latent representation $\mathbf{z}_i = f_\theta(\mathbf{X}_i)$. These latent features capture essential temporal and cross-sensor patterns, reducing noise and dimensionality compared to raw sensor signals. A classifier g_ϕ then maps the latent representation to an activity label y_i, i.e., $y_i = g_\phi(\mathbf{z}_i)$. Once trained, the activity recognition model performs real-time inference on streaming data, enabling timely detection of operational activities such as driving, idling, stopping, loading, and unloading as well as auxiliary operational activities that are not directly related to vehicle propulsion, such as electric power take-off (ePTO) usage.

In parallel, auxiliary energy consumption forecasting is formulated as a regression task. A predictor h_ψ maps both the raw sensor inputs $\mathbf{X}_i$ and the learned latent activity representations $\mathbf{z}_i$ to an estimated auxiliary energy usage $\hat{e}_i$, i.e., $\hat{e}_i = h_\psi(\mathbf{X}_i, \mathbf{z}_i)$. The forecasting model is initially trained offline on historical data to capture general energy consumption patterns, and then updated sequentially in an online continual learning setting. This allows the system to adapt to concept drift, evolving operational conditions, and newly introduced vehicle equipment, while maintaining robust and accurate energy predictions throughout vehicle deployment. By explicitly modeling both the mapping from raw features to targets and from latent features to targets, this framework leverages feature representations to improve prediction accuracy and interpretability for downstream tasks.

2.1 Research Questions

1. How can compact and meaningful latent embeddings be learned through offline training on historical multivariate CAN-bus time-series data, and to what extent do these embeddings support accurate real-time vehicle activity classification?

2. How can learned activity embeddings be integrated into auxiliary energy consumption forecasting models that combine offline training with online continual updates, and does incorporating these embeddings improve prediction accuracy compared to models using only raw sensor data?

3. How can online continual learning be applied to auxiliary energy consumption forecasting to adapt to evolving operational conditions, new equipment usage (e.g., electric power take-off systems), and driver- or season-specific patterns, while maintaining prediction stability and accuracy over time?

4. What design strategies enable an integrated system with offline-trained activity recognition and online-adaptive energy forecasting to remain scalable, efficient, and label-efficient, without requiring frequent retraining from scratch?

Accurate vehicle activity recognition and auxiliary energy forecasting are essential for improving operational efficiency, range, and fleet management of heavy-duty battery electric vehicles (BEVs). Offline-trained activity recognition models can generate compact and meaningful latent embeddings from multivariate sensor data, capturing key temporal and cross-sensor patterns that enable accurate real-time classification of operational states. These embeddings can then be leveraged to enhance auxiliary energy forecasting models, which combine offline training on historical data with online continual updates, allowing adaptation to evolving operational conditions, new equipment usage (e.g., ePTO systems), and driver- or season-specific patterns. By explicitly modeling the mapping from raw sensor inputs ($\mathbf{X}$) and latent activity representations ($\mathbf{z}$) to targets, the proposed framework aims to maintain high accuracy and robustness while minimizing reliance on extensive labeled datasets and frequent retraining, providing a scalable and efficient solution for long-term deployment in real-world heavy-duty BEVs.

2.2 Challenges

Vehicle activity recognition and auxiliary energy consumption modeling in heavy-duty battery electric vehicles pose several practical challenges. Although each operational task can be learned from a well-defined data distribution, gradual concept drift is likely to occur due to varying operating conditions, seasonal effects, and changes in vehicle usage or equipment, such as the introduction of electric power take-off systems. At the same time, obtaining labeled data for vehicle operational activities remains costly and impractical at fleet scale, limiting the applicability of fully supervised approaches. Sequential model updates further risk catastrophic forgetting, where previously learned activity patterns degrade as new tasks are introduced. In addition, vehicle behaviors exhibit complex temporal dependencies across multiple time scales, requiring models capable of capturing both short-term dynamics and longer-term patterns from multivariate sensor data. Finally, auxiliary energy consumption is tightly coupled with operational activities and contextual factors, motivating an integrated learning framework that jointly supports activity recognition and energy forecasting.

3 Proposed Solution Overview

Accurate recognition of vehicle activities and forecasting of auxiliary energy consumption in heavy-duty battery electric vehicles require a learning framework that effectively leverages sensor data while remaining scalable and robust to evolving conditions. In this work, activity recognition models are trained offline using labeled multivariate sensor data collected from the vehicle CAN bus. This allows the system to capture complex temporal dependencies across multiple time scales, including both short-term dynamics and longer-term patterns associated with operational states such as driving, idling, stopping, loading, and unloading.

The trained models generate compact and informative latent embeddings that summarize high-dimensional raw sensor data into structured representations. These embeddings reduce dimensionality, capture interdependencies among sensor signals, and create a semantically meaningful feature space that improves classification accuracy. They also provide interpretability, enabling domain experts to review and label clusters of vehi-

cle activities, and can be reused across multiple downstream tasks without retraining, enhancing scalability and efficiency for large-scale fleet deployment.

Building on these embeddings, auxiliary energy consumption forecasting is performed using a hybrid framework that combines offline training on historical energy data with online sequential updates as new measurements become available. This allows the models to adapt to changing operating conditions, seasonal variations, driver-specific behaviors, and newly introduced equipment such as electric power take-off (ePTO) systems, while maintaining stability and robustness over time. By explicitly modeling the mapping from raw sensor inputs (X) and latent activity representations (z) to energy consumption targets, the framework ensures accurate and scalable predictions across a fleet of vehicles.

Overall, this integrated approach addresses the dual challenges of efficient activity recognition and adaptive energy forecasting, providing a practical, interpretable, and reusable solution that supports intelligent, data-driven decision-making for fleet management and vehicle design.

4 Literature Review

A broad body of research has explored representation learning and time-series modeling for activity recognition and energy consumption prediction in intelligent transportation systems. Self-supervised and contrastive learning methods have gained particular attention in domains where labeled data are scarce and continuous adaptation to streaming sensor data is required.

Khaertdinov et al. [3] propose a contrastive self-supervised learning framework for sensor-based human activity recognition that learns robust representations from unlabeled multivariate time-series data. Their approach demonstrates strong performance in limited-label scenarios and highlights the effectiveness of self-supervised learning for extracting generalizable features from sensor streams. However, the method is evaluated in the context of human activity recognition and does not address vehicle-specific operational settings or online learning.

More recently, Chang et al. [1] introduce TimeDRL, a self-supervised framework for disentangled representation learning in multivariate time-series. By explicitly deriving timestamp-level and instance-level embeddings using predictive and contrastive pretext tasks, TimeDRL mitigates inductive biases introduced by data augmentation and achieves strong performance across forecasting and classification benchmarks.However, the framework is evaluated in an offline learning setting.

In the transportation domain, Liu et al. [4] introduce a distributional and spatial-temporal robust representation learning framework that integrates statistical, spatial, and temporal features from multimodal sensor data. Their model achieves state-of-the-art performance on transportation activity recognition benchmarks, emphasizing the importance of modeling temporal dependencies and spatial relationships. An ablation study further highlights the contribution of each component to the model's robustness.

Several studies have applied unsupervised and clustering-based methods to heavy-duty vehicles to analyze driving behavior and its relationship with energy consumption. Fang et al. [2] develop an offline–online framework for driving behavior segmentation using

Toeplitz Inverse Covariance Clustering (TICC). By relating identified driving behaviors to fuel consumption and contextual factors, their approach provides interpretable insights and eco-driving feedback.

Energy consumption forecasting for heavy-duty battery electric vehicles has been investigated using supervised machine learning models. Wang et al. [6] evaluate several learning-based approaches for predicting auxiliary energy consumption in heavy-duty BEVs. Their results demonstrate the suitability of temporal models for capturing operational dependencies but rely on offline training and assume stationary data distributions.

In summary, existing literature shows that self-supervised representation learning can effectively model sensor-based time series, disentangled representations improve generalization in multivariate time-series tasks, clustering methods enable interpretable behavior recognition, and machine learning approaches can achieve accurate energy consumption forecasting. However, few studies integrate online activity recognition with continual, label-efficient energy forecasting for heavy-duty battery electric vehicles. Addressing this gap motivates the contributions of this thesis.

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5 Contributions of the Thesis

This thesis makes two main contributions. First, it develops a labeled dataset for heavy-duty battery electric trucks, including activities related to electric power take-off (ePTO) systems. The dataset is constructed from multivariate CAN bus time-series data and labeled using a combination of subsequence clustering methods and domain expert knowledge, enabling reliable identification of vehicle operational activities. Second, the thesis presents a systematic comparison of state-of-the-art activity recognition methods for electric trucks, evaluating their performance on real-world operational data and highlighting their strengths and limitations in practical deployment scenarios.

5.1 Novelty

The novelty of this thesis lies in the development of an integrated framework for heavy-duty battery electric vehicles that combines offline-trained activity recognition with adaptive auxiliary energy consumption forecasting. The proposed approach leverages self-supervised learning to generate compact and meaningful latent embeddings from multivariate CAN-bus sensor data, capturing complex temporal patterns of vehicle operations. These embeddings provide interpretable representations that can be reviewed and labeled by domain experts, and can be reused across multiple downstream tasks without retraining, ensuring scalability and consistency in activity recognition.

Building on these embeddings, a hybrid energy forecasting model combines offline training on historical energy data with online sequential updates. This allows the system to adapt to evolving operational conditions, driver-specific behaviors, and newly introduced equipment such as ePTO systems, while maintaining robust and accurate predictions. Additionally, this work contributes a novel labeled dataset of heavy-duty electric vehicle operational activities, constructed using embedding-based clustering combined

with expert knowledge. Together, these contributions provide a scalable, interpretable, and label-efficient framework for intelligent fleet management, advancing the state-of-the-art in vehicle activity recognition and energy consumption forecasting.

6 Project Methodology and Thesis planning

The project aims to improve the operational efficiency of heavy-duty battery electric vehicles (BEVs) by recognizing vehicle activities and forecasting auxiliary energy consumption. Multivariate time-series data are collected from CAN bus sensors, including speed, acceleration, battery state, ambient conditions, and other operational signals. The raw data are preprocessed to handle missing values through piecewise interpolation, remove outliers using statistical criteria, and normalize features to ensure consistency and reliability for subsequent modeling.

Activity recognition is performed using an offline self-supervised contrastive learning framework, which generates compact and informative embeddings of vehicle behaviors from multivariate CAN-bus sensor data. These embeddings capture meaningful temporal and behavioral patterns without requiring extensive labeled datasets, reducing labeling effort while providing robust and interpretable representations. Domain experts assign semantic labels to the embeddings, enabling accurate recognition of operational activities in real-world conditions. The embeddings are reusable across downstream tasks, supporting consistent monitoring and analysis without retraining the activity recognition model.

The recognized activities are incorporated into a hybrid forecasting model for auxiliary energy consumption. This model combines offline training on historical trip data with online continual updates, allowing adaptation to evolving driver behavior, vehicle operation, seasonal variations, and newly introduced equipment. By continuously updating the embeddings and forecasting model, the system mitigates concept drift and maintains high predictive performance in dynamic operational conditions.

The integrated framework links activity recognition with energy forecasting, enabling fleet managers to monitor energy-intensive behaviors and receive actionable recommendations for efficiency improvements. Evaluation is performed by measuring activity recognition accuracy, comparing energy consumption predictions against baseline models, and assessing the system's adaptability to new trips and changing conditions. Overall, this end-to-end methodology provides a data-driven solution for intelligent fleet management, optimizing vehicle performance and energy usage through continuous learning from streaming sensor data.

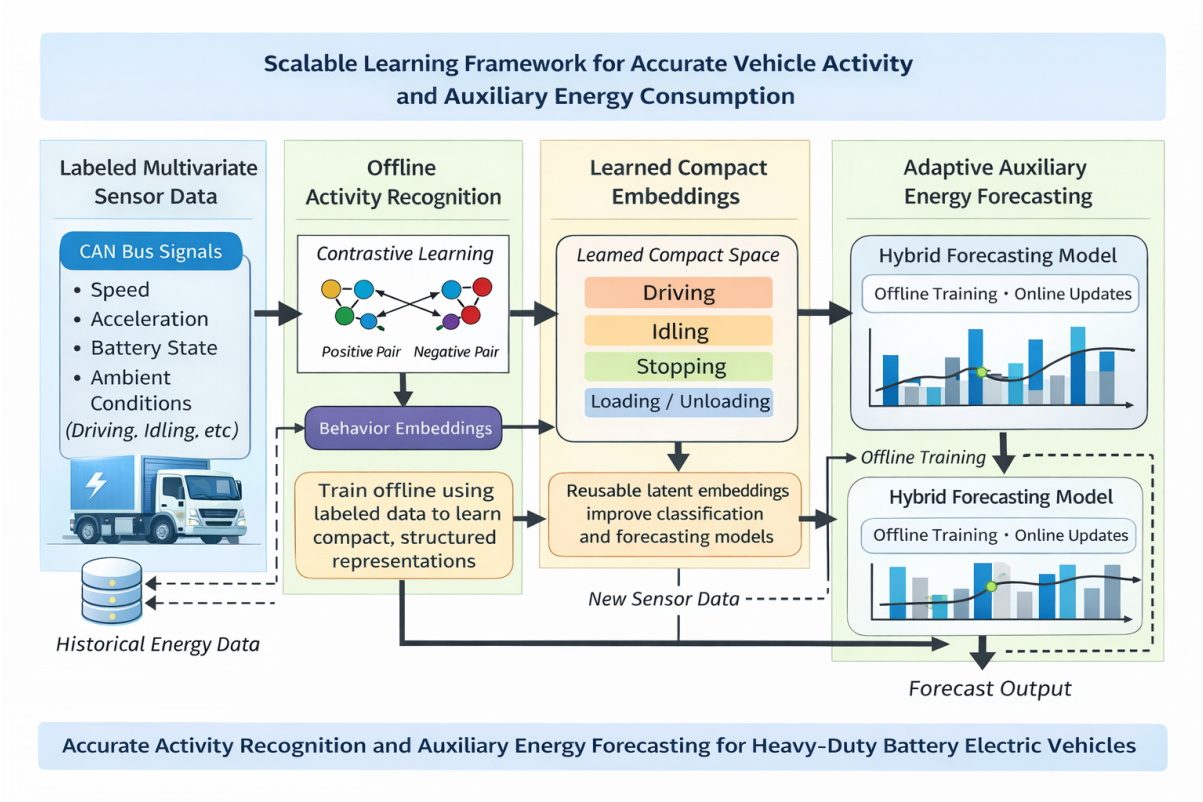

Figure 1: Model

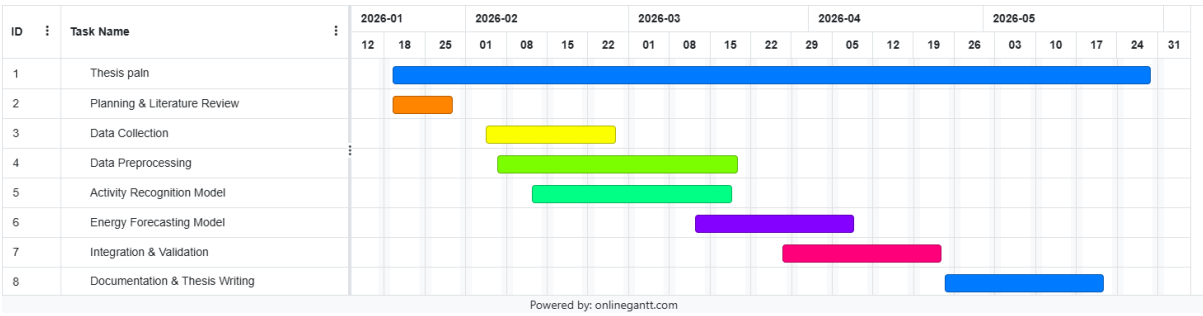

Figure 2: Gantt Chart

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